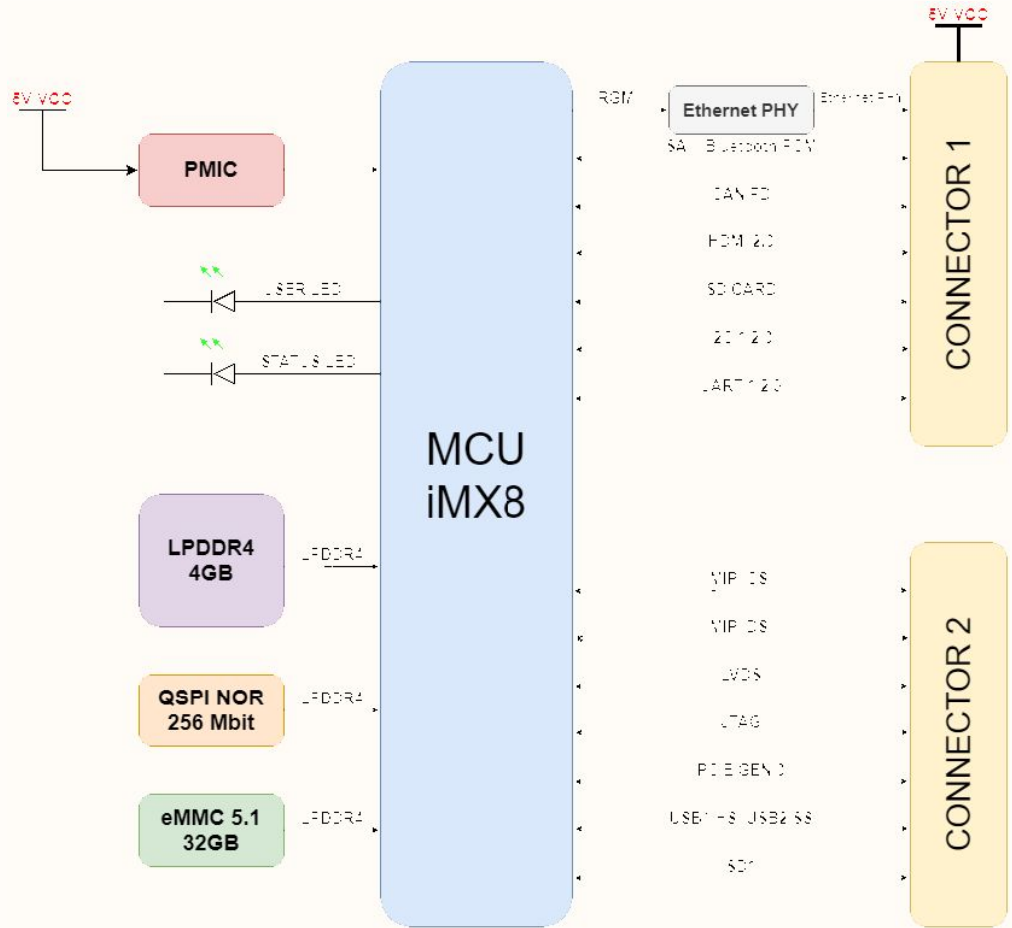


BLOCK DIAGRAM



A



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V1I1

PRELIMINARY

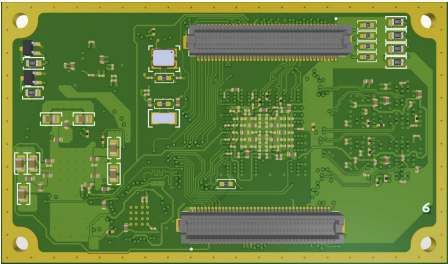
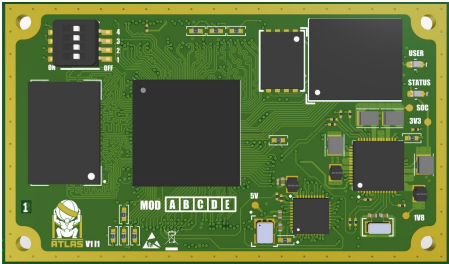
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9	CPU - HDMI, CSI, DSI, LVDS	19		29		39	
10	LPDDR4	20		30		40	

EXAMPLE TEXT FOR
INFORMATIONAL DESIGN
NOTES

EXAMPLE TEXT FOR
CRITICAL DESIGN
NOTES

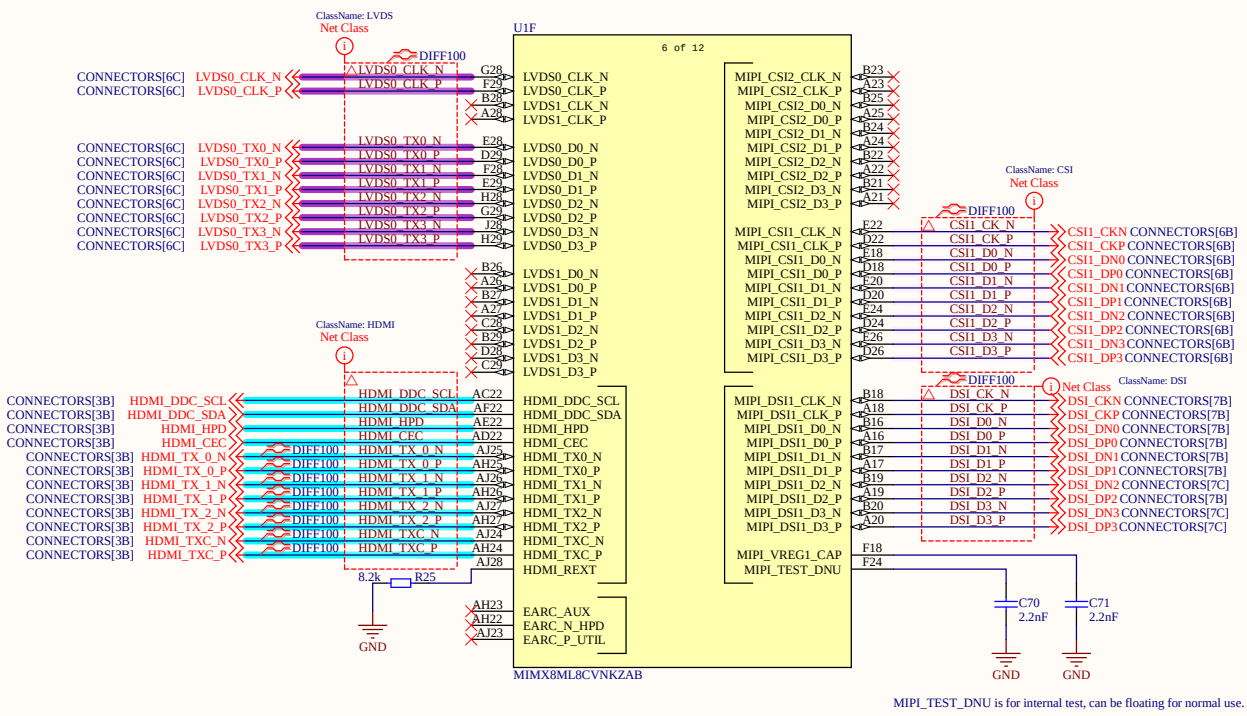
EXAMPLE TEXT FOR
CAUTIONARY DESIGN
NOTES

EXAMPLE TEXT FOR
CRITICAL LAYOUT
GUIDELINES

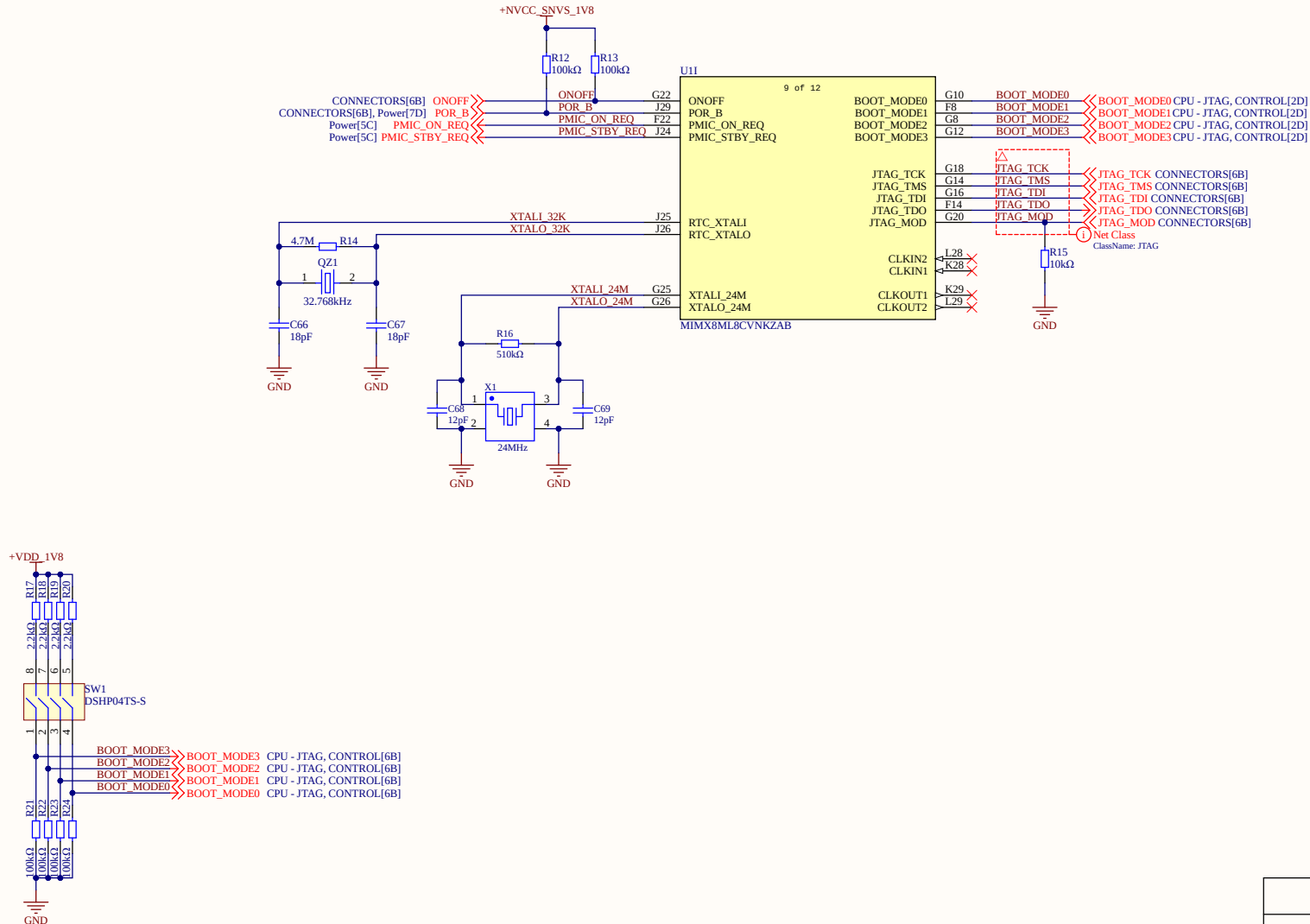


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CPU - HDMI, LVDS, CSI, DSI



CPU - JTAG, CONTROL



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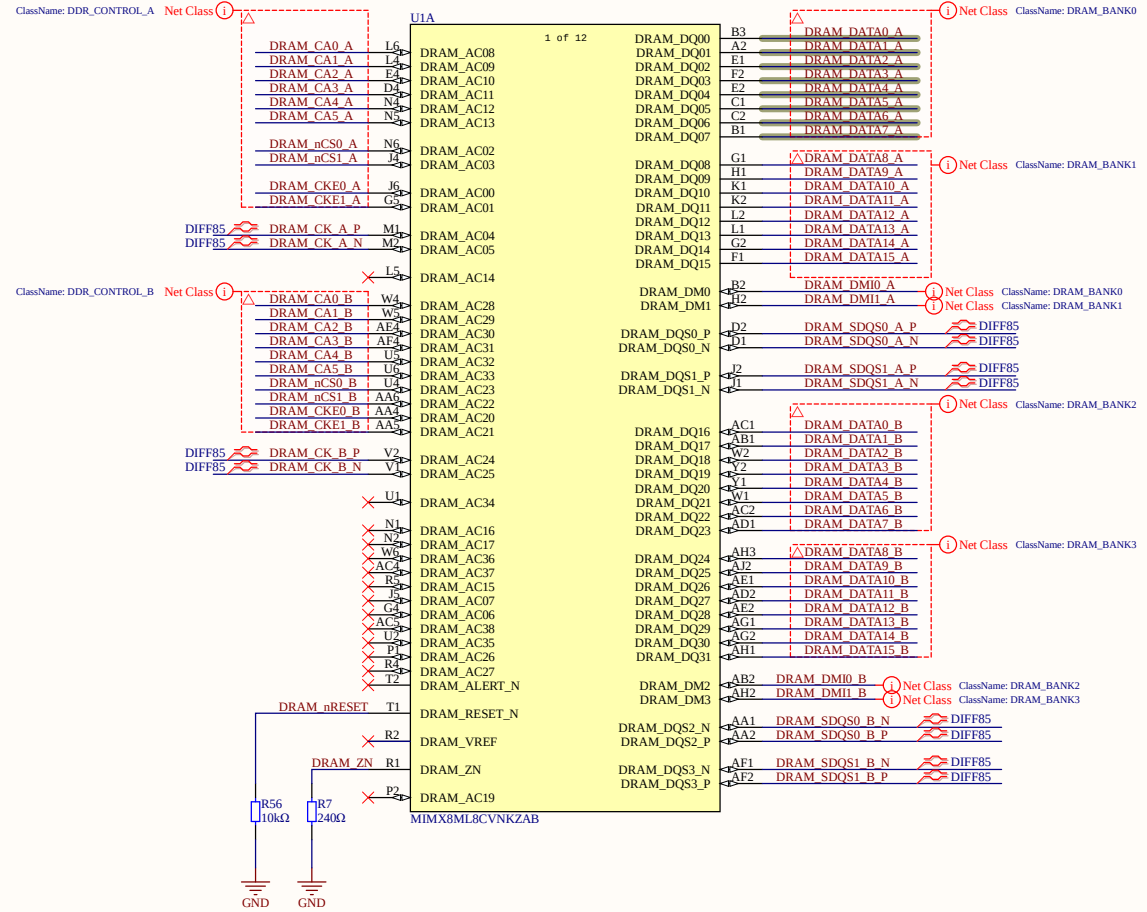
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CPU - DDR4

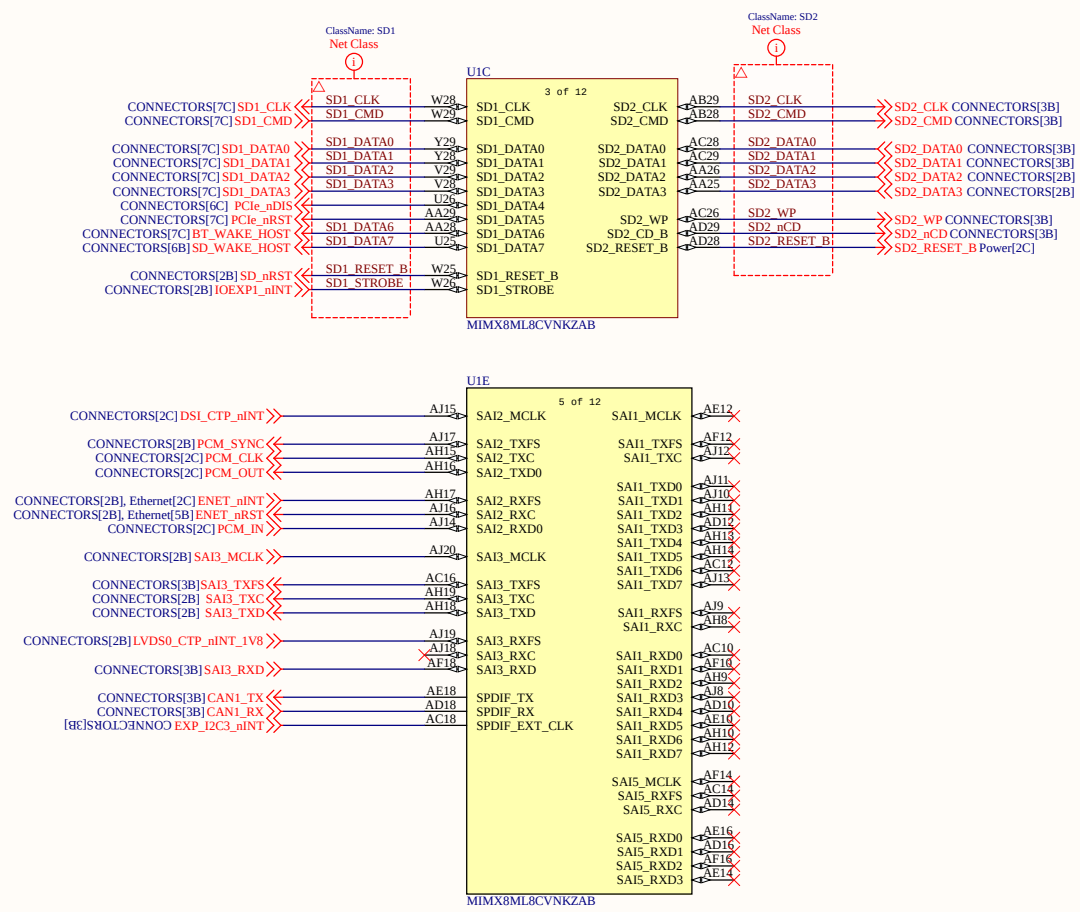


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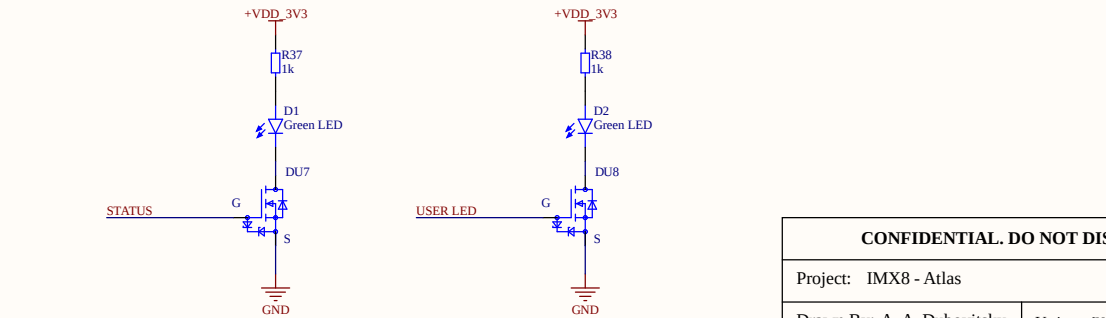


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CPU - SD, SAI

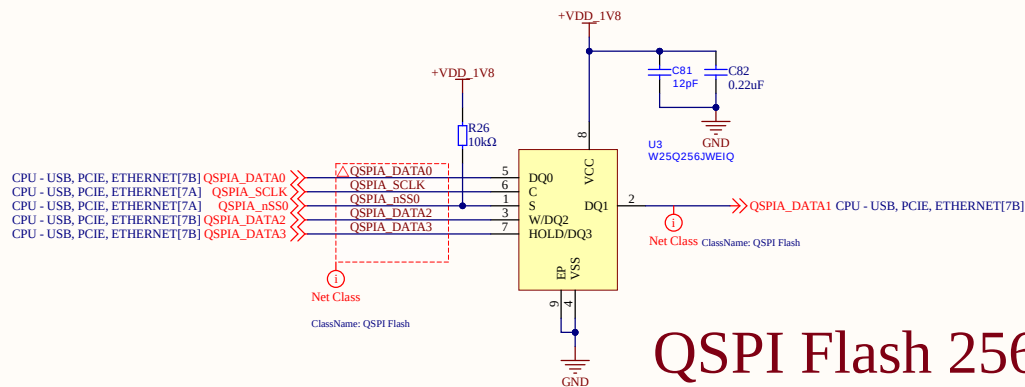
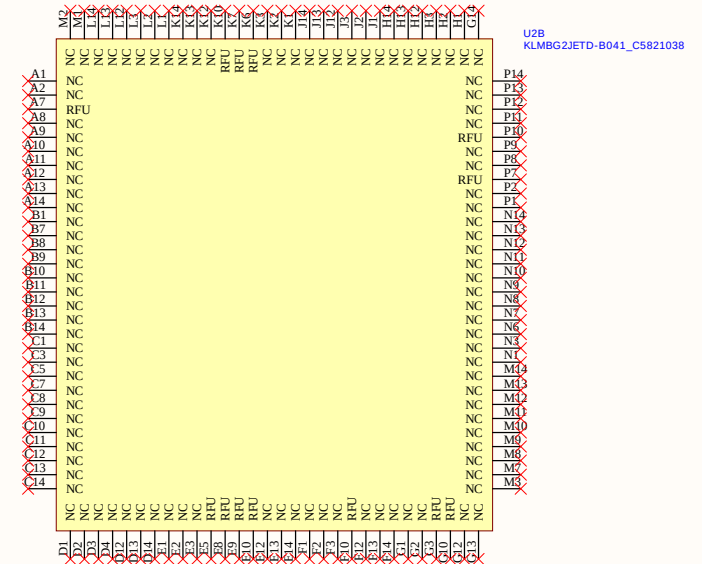


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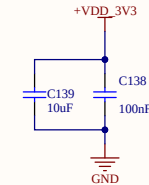
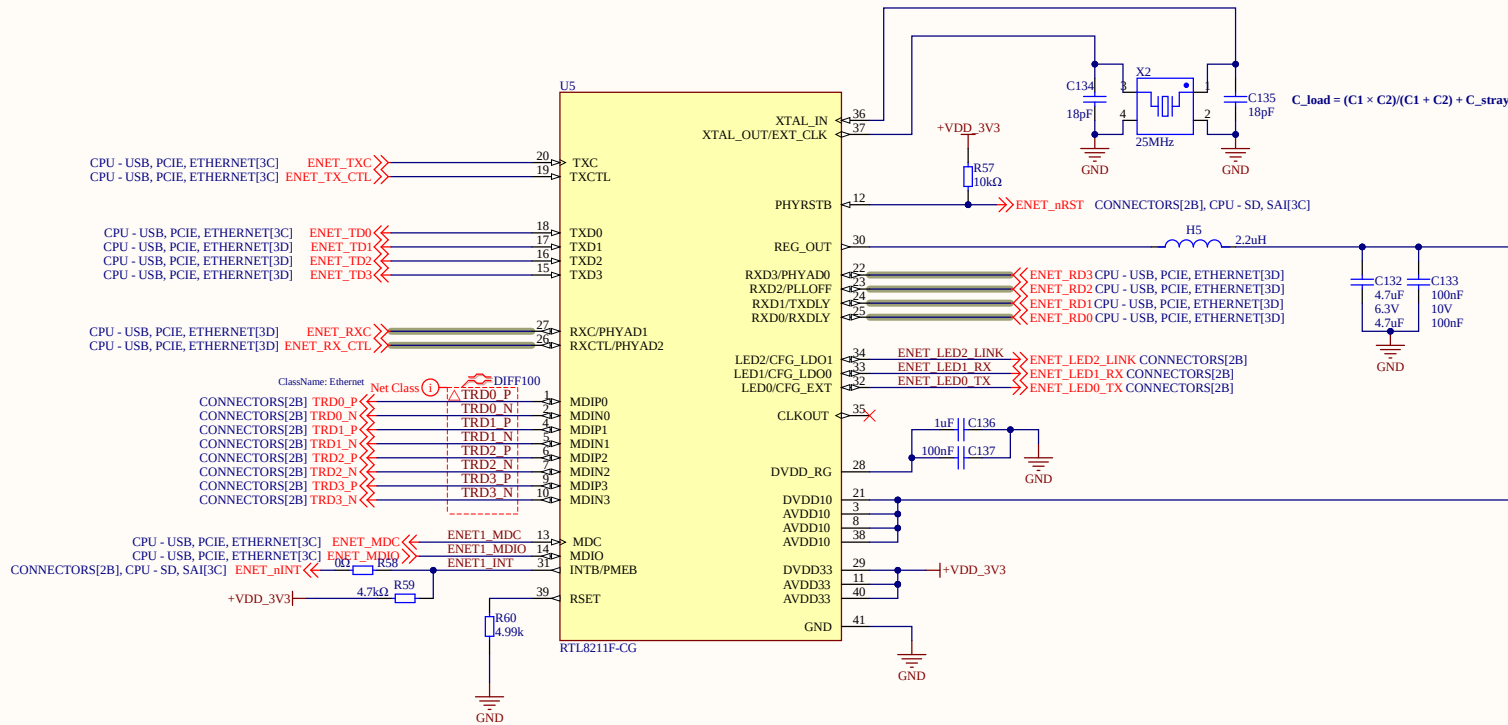
MMC 32GB



QSPI Flash 256 Mbit

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ETHERNET



9.1.1. PCB Layout

- The input 3.3V power trace connected to DVDD33 must be wider than 40mils
- The bulk de-coupling capacitors (Cin1 and Cin2) must be placed within 200mils (0.5cm) of DVDD33 to prevent input voltage overshoot
- The output power trace out of **REG_OUT** must be wider than 60mils
- Lx (2.2uH/4.7uH) must be kept within 200mils (0.5cm) of REG_OUT
- Cout1 and Cout2 must be kept within 200mils (0.5cm) of Lx to ensure stable output power and better power efficiency
- For switching regulator stability, the capacitor Cout1 and Cout2 must be a ceramic (X5R) capacitor. Cin1 and Cin2 are recommended to be ceramic capacitors
- Place Lx and Cin1 on the same layer as the RTL8211F(I). Do not use vias on DVDD33 and REG_OUT traces

Note: Violation of the above rules will damage the IC.

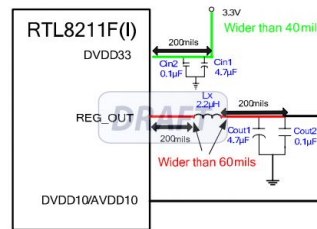


Figure 10. Switching Regulator

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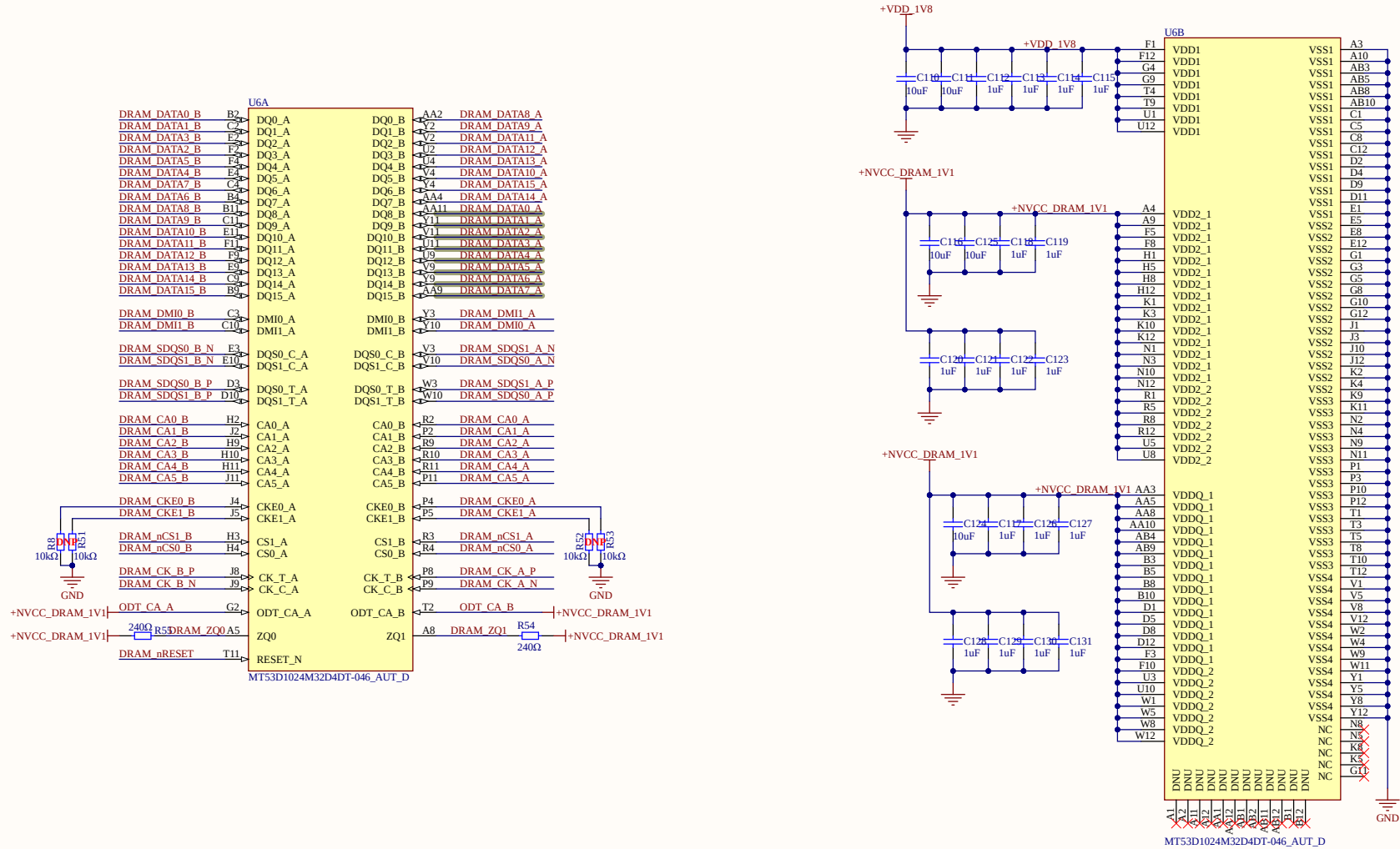
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LPDDR4



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Variant: [No Variations]

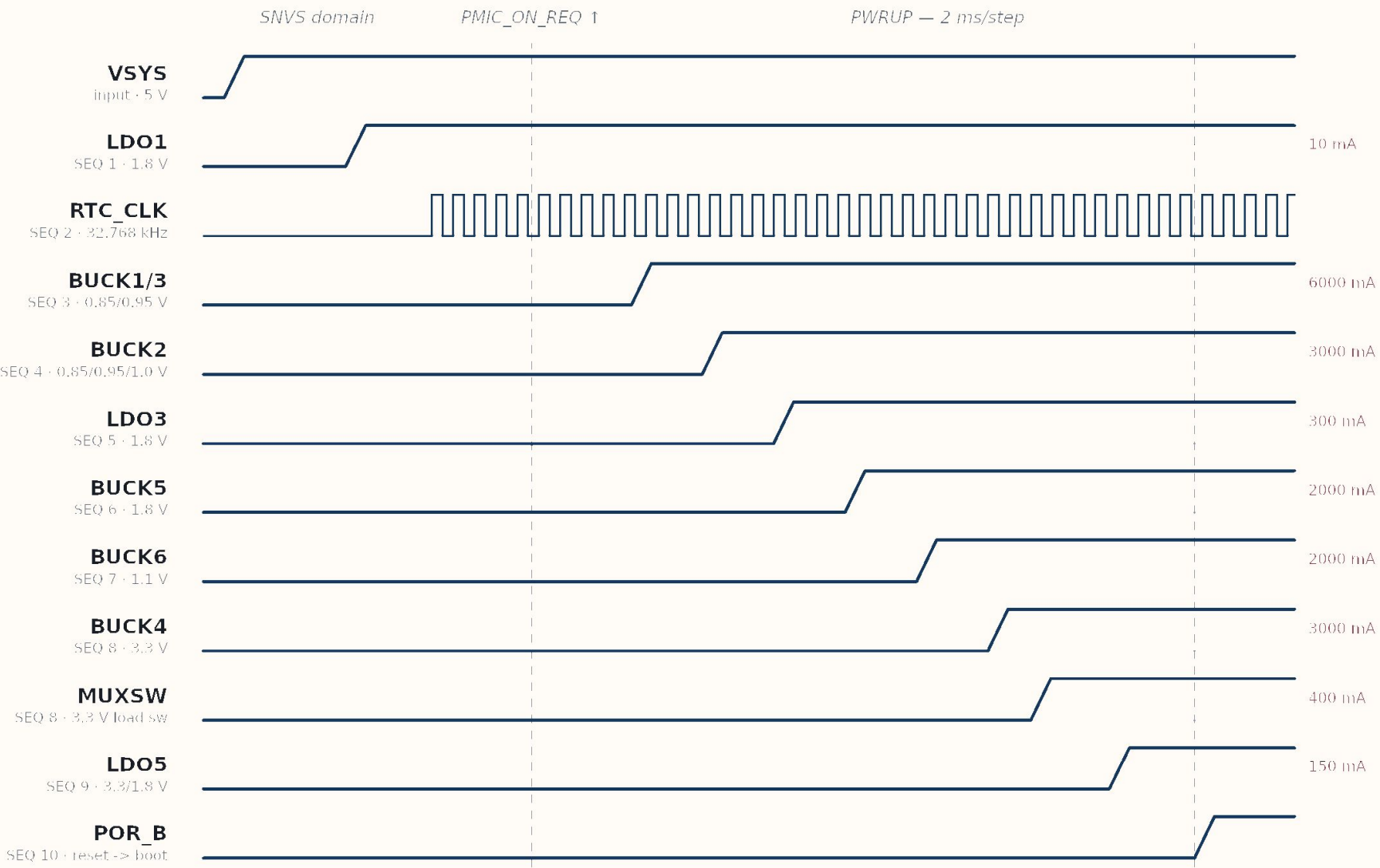
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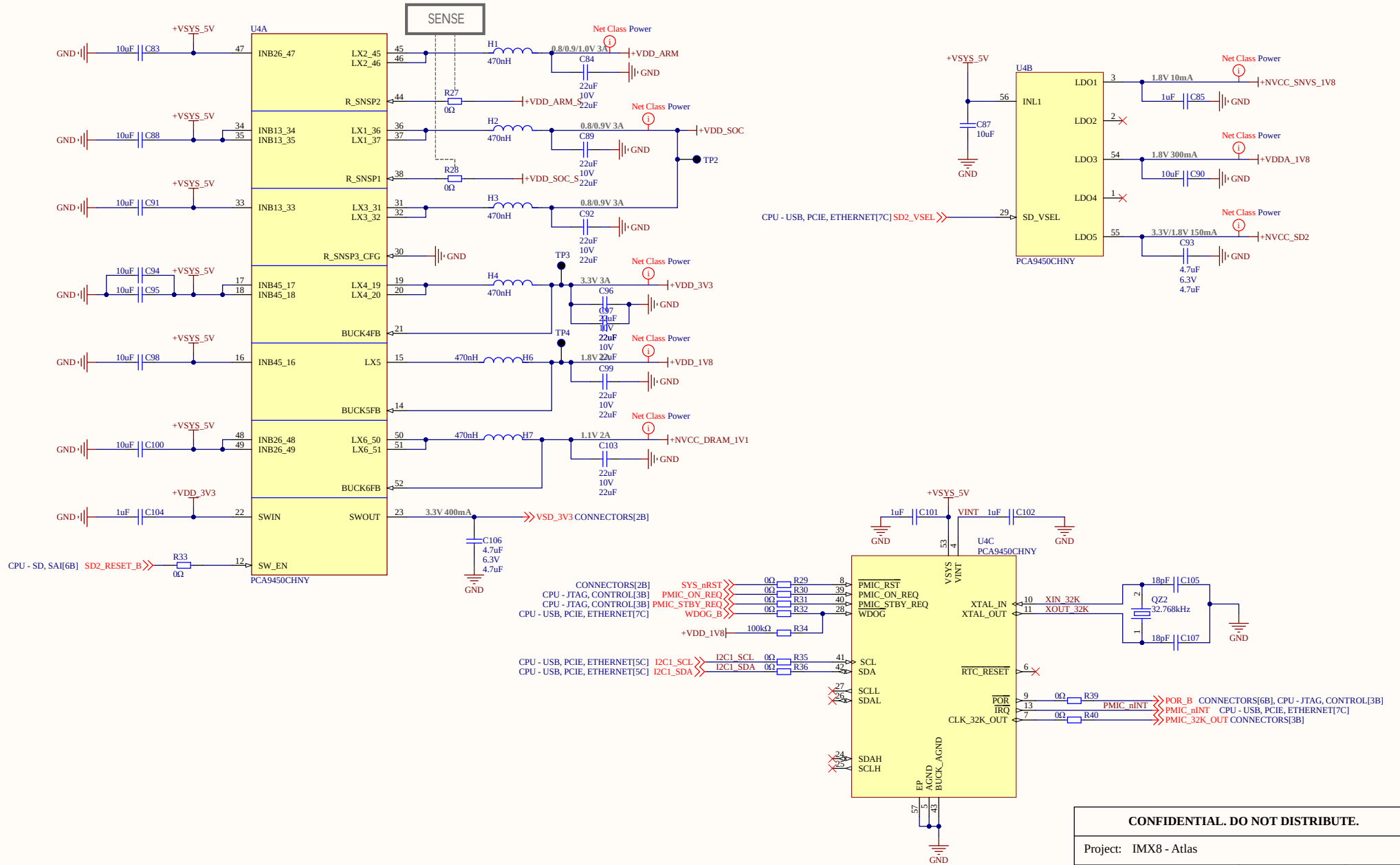
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POWER SEQUENCING



POWER



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Variant: [No Variations]	
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A.A.Д.

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B								B
C								C
D								D
1	2	3	4	5	6	7	8	

DATE	COMMENT
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Drawn By: A. A. Dubovitsky	Variant: [No Variations]	Version: VIII
<u>A.A.A.</u>	Date: 28/08/2026	Page 16 of 16

DATE	COMMENT
28-AUG-2026	DRAFT FOR INITIAL REVIEW

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